FLEET TEAM DIGEST

737MAX-FTD-34-19001

Issue Title : Honeywell IMMR Intermittent GPS Unavailability

Airplane Model	ATA	Minor Model(s)
737MAX	3458-00	-7, -8, -9, -BBJ

Other Model(s)	ECCN
737NG	9E991

Originated Date	Last Revised Date	Created On
02/15/2019	10/26/2019	10/09/2019

Estimated Completion Date	Next Update Date

Revision Description

- Noted additional concerns being addressed by software update.

- SB has been released, provided SB number and release date.

Applicability

737NG and 737MAX airplanes utilizing Honeywell Intergrated Multi-Mode Receiver (IMMR), PN: 69002600-0101.

Description

This FTD provides information regarding the intermittent GPS-L INVALID and/or GPS-R INVALID message shown on the Control Display Unit (CDU), along with dashes shown on the corresponding clock, while operating in the Asia/Pacific Region. The fail indicator light on the Air Traffic Control (ATC) panel may also illuminate.

Background

Boeing received reports from operators in the Asia/Pacific region, of the GPS-L INVALID or GPS-R INVALID message being displayed on the CDU scratchpad, along with dashes on the corresponding clock. This message originates from the Flight Management Computer (FMC) which monitors GPS data. If the FMC declares GPS data invalid for 30 seconds while airborne, or 10 minutes while on the ground, this message will be displayed. The longer time requirement on the ground is to account for the GPS to locate satellites after initialization. This condition usually occurs when the GPS functionality in the IMMR does not have sufficient satellite signal and generally recovers automatically within 10 minutes. Additionally, ADS-B data, which is transmitted by the ATC transponder, is reliant on GPS position and therefore, the fail light on the ATC panel may also illuminate.

These symptoms are different from a hard fault of the IMMR unit or GPS antenna. The IMMR unit includes built in functionality to monitor for internal faults and GPS antenna faults. When a hard fault is detected for a single IMMR unit or GPS antenna, and the system status annunciator panel (recall) button is pressed, the “GPS” light on the IRS Mode Select panel, “IRS” light on the system status indicator panel, and the left and right master caution lights will all illuminate. When a hard fault is detected for both IMMR units simultaneously the same indications will be observed without pressing the recall button. Correlated maintenance messages

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should be logged when a hard fault is declared. The affected GPS system is usually no longer operable until hardware is physically replaced.

Boeing's initial assessment, is that the GPS-L INVALID or GPS-R INVALID message displayed on the CDU scratchpad is considered to be a nuisance condition and has no safety impact. However, Boeing is interested in understanding the root cause of these events and is currently investigating the IMMR as the cause of these events.

Status

Boeing has been working with Honeywell to determine the root cause of the invalid GPS data. IMMR units that have experienced an invalid GPS event were sent to Honeywell for extraction of internal memory logs. Honeywell has extracted the log data and performed the necessary analysis to determine root cause. Honeywell laboratory testing has been successful in replicating the experienced behavior and has found the cause to be attributed to signals originating from a recently introduced Satellite-based Augmentation System (SBAS) satellite in the Asia/Pacific Region. The signals being received from this SBAS satellite have been seen to alter a time related parameter, which in turn affects the calculation of GPS position, within the IMMR. Based on these results, Honeywell engineering has indicated a software update to the IMMR unit is necessary to rectify the condition.

Upon further analysis of the IMMR software code, additional findings were discovered and detailed in Honeywell SIL, D201906000005 Rev. 0, which relate to delayed annunciation of GNSS receiver faults. As noted in the Honeywell SIL, this delayed annunciation is not consistent with requirements defined by TSO-C145d, TSO-C146d, or TSO-C161a. This deficiency is resolved by installing a new version of IMMR software. Additional concerns affecting how the data is written to memory (MRAM), and satellite acquisition near the horizon are also addressed by this software.

Boeing and Honeywell have completed certification for the updated IMMR software and SB 737-34-3483 has been released on 21-OCT-2019 to allow retrofit installation.

Interim Action

No Interim Action Required.

Final Action

Final action consists of the certifying and releasing an updated IMMR software, which resolves these events, for installation on the fleet.

Milestones

Note: Target dates are based upon currently available status and have risk of future revision.

Root Cause Established: 19-MAR-2019 (complete)

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Service Bulletin Available: 21-OCT-2019 (complete)

Operator Action

When one GPS receiver data is declared invalid and the appropriate message is displayed, the FMC then defaults to the remaining GPS receiver data for position update, and redundancy of the system is lost. When both GPS receiver data are declared invalid, the FMC reverts to radio updating if available, to continue navigating, provided the airspace requirement does not require GPS equipment. In the case that radio navigation facilities are not available, the FMC uses the IRS to produce a navigation solution.

During times of invalid GPS, flight crews should be aware of their RNP requirement and ensure their ANP is adequate. The RNP Capabilities document provides the equipment list required for each airspace. Each operator needs to develop guidance materials for their crews regarding degradation or loss of navigation capabilities in accordance with their regulatory authorities.

For aircraft equipped with ADSB-Out Transponders, if the left IMMR loses GPS and ATC transponder 1 is selected, the ATC Control Panel fail light will illuminate. The same is true if the right IMMR loses GPS and ATC transponder 2 is selected. If the offside IMMR did not lose GPS, the offside ATC transponder can be selected to continue transmitting ADS-B -Out position data.

Operators are also recommended to inform maintenance crews to be aware that if the ATC transponder, IMMR, or GPS antenna hardware has a hard fault, then a correlated maintenance message would also be present. FIM troubleshooting and/or AMM tasks should be performed as necessary to rectify the maintenance message.

Supplier Information

HONEYWELL INTERNATIONAL INC.

15001 NE 36TH ST

REDMOND, WA

98052-5317

Reference Categories

Reference Type	Reference Number	Revision Number	Reference Date
Service Bulletin (SB)	737-34-3483	00	10/21/2019
Service Information Letter (SIL)	Honeywell - D201902000041	1	03/19/2019
Service Information Letter (SIL)	Honeywell - D201906000005	0	06/04/2019

Parts List

Part Name	Boeing Part Number	Supplier Name	Supplier Number
Integrated Multi-Mode Receiver		Honeywell International Inc.	69002600-0101

Related Categories

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- Component Issues : Issues that occur during component maintenance or have an impact on component reliability

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